Table A1

Table A1 provides the complete enumeration of the 106 descriptors comprising the AutoAIM Custom featurizer. These descriptors implement the elemental-statistical approach conceptually equivalent to Magpie [14]: 13 elemental properties are encoded through statistical moments. All elemental data are drawn from standard tabulated sources embedded within the AutoAIM software.

Table A1 (Part 1/5). Elemental Properties (40 descriptors = 5 properties x 8 statistics)

Category	Descriptor Name	Definition	Source
Elemental	comp_atomic_number_mean	Weighted mean: $\bar{x} = \Sigma\ fi \cdot atomic_numberi$. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_std	Weighted std: $\sigma = \sqrt{[\Sigma(xi - \bar{x})^2 \cdot fi]}$. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_min	Minimum atomic number among constituent elements. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_max	Maximum atomic number among constituent elements. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_range	Range: max(atomic_number) - min(atomic_number). Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_median	Weighted median of atomic number. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_p25	25th percentile (Q1) of atomic number. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_p75	75th percentile (Q3) of atomic number. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_mean	Weighted mean: $\bar{x} = \Sigma\ fi \cdot atomic_massi$. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Elemental	comp_atomic_mass_std	Weighted std: $\sigma = \sqrt{[\Sigma(xi - \bar{x})^2 \cdot fi]}$. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_min	Minimum atomic mass among constituent elements. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_max	Maximum atomic mass among constituent elements. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_range	Range: max(atomic_mass) - min(atomic_mass). Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_median	Weighted median of atomic mass. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_p25	25th percentile (Q1) of atomic mass. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_p75	75th percentile (Q3) of atomic mass. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_group_mean	Weighted mean: $\bar{x} = \Sigma\ fi \cdot groupi$. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_group_std	Weighted std: $\sigma = \sqrt{[\Sigma(xi - \bar{x})^2 \cdot fi]}$. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_group_min	Minimum group among constituent elements. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_group_max	Maximum group among constituent elements. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_group_range	Range: max(group) - min(group). Periodic table column	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Elemental	comp_group_median	Weighted median of group. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_group_p25	25th percentile (Q1) of group. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_group_p75	75th percentile (Q3) of group. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_period_mean	Weighted mean: $\bar{x} = \Sigma\ f_i \cdot \text{periodi}$. Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_period_std	Weighted std: $\sigma = \sqrt{[\Sigma(x_i - \bar{x})^2 \cdot f_i]}$. Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_period_min	Minimum period among constituent elements. Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_period_max	Maximum period among constituent elements. Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_period_range	Range: max(period) - min(period). Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_period_median	Weighted median of period. Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_period_p25	25th percentile (Q1) of period. Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_period_p75	75th percentile (Q3) of period. Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_mean	Weighted mean: $\bar{x} = \Sigma\ f_i \cdot \text{atomic_radius}_i$. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Elemental	comp_atomic_radius_std	Weighted std: $\sigma = \sqrt{[\Sigma(x_i - \bar{x})^2 \cdot f_i]}$. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_min	Minimum atomic radius among constituent elements. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_max	Maximum atomic radius among constituent elements. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_range	Range: max(atomic_radius) - min(atomic_radius). Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_median	Weighted median of atomic radius. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_p25	25th percentile (Q1) of atomic radius. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_p75	75th percentile (Q3) of atomic radius. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)

Table A1 (Part 2/5). Statistical Moments (24 descriptors = 3 properties x 8 statistics)

Category	Descriptor Name	Definition	Source
Statistical	comp_covalent_radius_mean	Weighted mean: $\bar{x} = \Sigma f_i \cdot \text{covalent_radius}_i$. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_covalent_radius_std	Weighted std: $\sigma = \sqrt{[\Sigma(x_i - \bar{x})^2 \cdot f_i]}$. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_covalent_radius_min	Minimum covalent radius among constituent elements. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_covalent_radius_max	Maximum covalent radius among constituent elements. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Statistical	comp_covalent_radius_range	Range: max(covalent_radius) - min(covalent_radius). Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_covalent_radius_median	Weighted median of covalent radius. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_covalent_radius_p25	25th percentile (Q1) of covalent radius. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_covalent_radius_p75	75th percentile (Q3) of covalent radius. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_mean	Weighted mean: $\bar{x} = \Sigma\ f_i \cdot \text{melting_point}_i$. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_std	Weighted std: $\sigma = \sqrt{[\Sigma(x_i - \bar{x})^2 \cdot f_i]}$. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_min	Minimum melting point among constituent elements. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_max	Maximum melting point among constituent elements. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_range	Range: max(melting_point) - min(melting_point). Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_median	Weighted median of melting point. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_p25	25th percentile (Q1) of melting point. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_p75	75th percentile (Q3) of melting point. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_mean	Weighted mean: $\bar{x} = \Sigma\ f_i \cdot \text{boiling_point}_i$. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_std	Weighted std: $\sigma = \sqrt{[\Sigma(x_i - \bar{x})^2 \cdot f_i]}$. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_min	Minimum boiling point among constituent elements. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_max	Maximum boiling point among constituent elements. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Statistical	comp_boiling_point_range	Range: max(boiling_point) - min(boiling_point). Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_median	Weighted median of boiling point. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_p25	25th percentile (Q1) of boiling point. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_p75	75th percentile (Q3) of boiling point. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)

Table A1 (Part 3/5). Electronic Structure (18 descriptors: electronegativity 8 + ionization_energy 8 + electron_affinity 2)

Category	Descriptor Name	Definition	Source
Electronic	comp_electronegativity_mean	Weighted mean: $\bar{x} = \Sigma$ fi·electronegativityi. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_electronegativity_std	Weighted std: $\sigma = \sqrt{[\Sigma(xi - \bar{x})^2 \cdot fi]}$. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_electronegativity_min	Minimum electronegativity among constituent elements. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_electronegativity_max	Maximum electronegativity among constituent elements. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_electronegativity_range	Range: max(electronegativity) - min(electronegativity). Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_electronegativity_median	Weighted median of electronegativity. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_electronegativity_p25	25th percentile (Q1) of electronegativity. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Electronic	comp_electronegativity_p75	75th percentile (Q3) of electronegativity. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_mean	Weighted mean: $\bar{x} = \Sigma$ fi·ionization_energyi. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_std	Weighted std: $\sigma = \sqrt{[\Sigma(xi - \bar{x})^2 \cdot fi]}$. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_min	Minimum ionization energy among constituent elements. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_max	Maximum ionization energy among constituent elements. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_range	Range: max(ionization_energy) - min(ionization_energy). First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_median	Weighted median of ionization energy. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_p25	25th percentile (Q1) of ionization energy. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_p75	75th percentile (Q3) of ionization energy. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_electron_affinity_mean	Weighted mean: $\bar{x} = \Sigma$ fi·electron_affinityi. Electron affinity (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_electron_affinity_std	Weighted std: $\sigma = \sqrt{[\Sigma(xi - \bar{x})^2 \cdot fi]}$. Electron affinity (kJ/mol)	Tabulated elemental data (AutoAIM)

Table A1 (Part 4/5). Thermodynamic Properties (12 descriptors: thermal_conductivity 8 + density 4)

Category	Descriptor Name	Definition	Source
Thermodynamic	comp_thermal_conductivity_mean	Weighted mean: $\bar{x} = \Sigma$ fi·thermal_conductivityi. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_std	Weighted std: $\sigma = \sqrt{[\Sigma(xi - \bar{x})^2 \cdot fi]}$. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_min	Minimum thermal conductivity among constituent elements. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_max	Maximum thermal conductivity among constituent elements. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_range	Range: max(thermal_conductivity) - min(thermal_conductivity). Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_median	Weighted median of thermal conductivity. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_p25	25th percentile (Q1) of thermal conductivity. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_p75	75th percentile (Q3) of thermal conductivity. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_density_mean	Weighted mean: $\bar{x} = \Sigma$ fi·densityi. Mass density at STP (g/cm³)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_density_std	Weighted std: $\sigma = \sqrt{[\Sigma(xi - \bar{x})^2 \cdot fi]}$. Mass density at STP (g/cm³)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_density_min	Minimum density among constituent elements. Mass density at STP (g/cm³)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_density_max	Maximum density among constituent elements. Mass density at STP (g/cm³)	Tabulated elemental data (AutoAIM)

Table A1 (Part 5/5). Composition Ratios (12 descriptors)

Category	Descriptor Name	Definition	Source
Composition	comp_n_elements	Number of unique constituent elements in the chemical formula	Derived from composition
Composition	comp_fractional_entropy	Shannon compositional entropy: H = -Σ fi·ln(fi)	Derived from composition
Composition	comp_ionization_energy_p25	25th percentile (Q1) of ionization energy. First ionization energy (kJ/mol)	Tabulated elemental data
Composition	comp_ionization_energy_p75	75th percentile (Q3) of ionization energy. First ionization energy (kJ/mol)	Tabulated elemental data
Composition	comp_electron_affinity_range	Range: max(electron affinity) - min(electron affinity). Electron affinity (kJ/mol)	Tabulated elemental data
Composition	comp_electron_affinity_median	Weighted median of electron affinity. Electron affinity (kJ/mol)	Tabulated elemental data
Composition	comp_electron_affinity_p25	25th percentile (Q1) of electron affinity. Electron affinity (kJ/mol)	Tabulated elemental data
Composition	comp_electron_affinity_p75	75th percentile (Q3) of electron affinity. Electron affinity (kJ/mol)	Tabulated elemental data
Composition	comp_density_range	Range: max(density) - min(density). Mass density at STP (g/cm³)	Tabulated elemental data
Composition	comp_density_median	Weighted median of density. Mass density at STP (g/cm³)	Tabulated elemental data
Composition	comp_density_p25	25th percentile (Q1) of density. Mass density at STP (g/cm³)	Tabulated elemental data
Composition	comp_density_p75	75th percentile (Q3) of density. Mass density at STP (g/cm³)	Tabulated elemental data